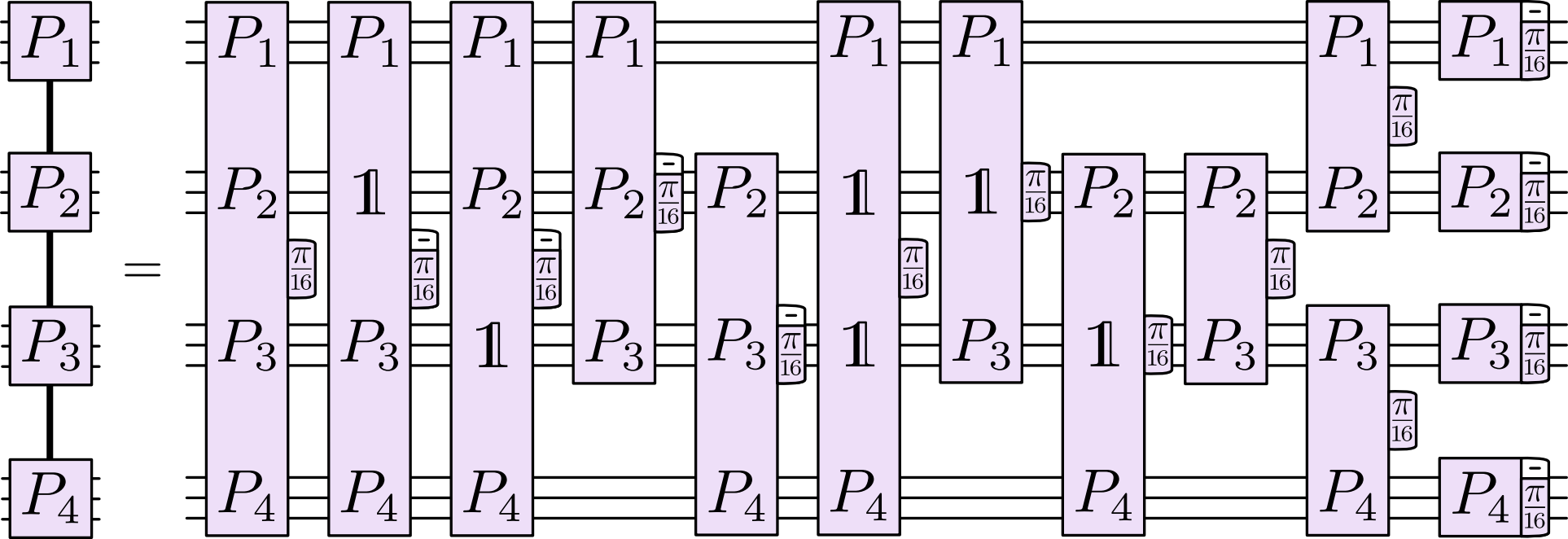
P_1
P_2
P_3
P_4
=
$\mathbb{1}$
$\frac{\pi}{16}$
$-\frac{\pi}{16}$